

AI & Gen AI

Primer

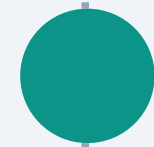
Foundations of Agentic AI & Generative Intelligence

Priya Dharsini

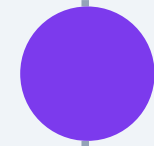
Session Roadmap



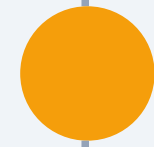
Classical ML vs Deep Learning vs Gen AI



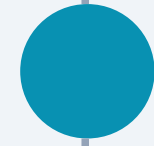
Transformer Architecture & Timeline



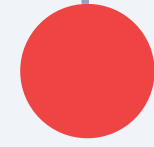
Diffusion vs Autoregressive Models



Open vs Closed Licences



Hugging Face Model Hub



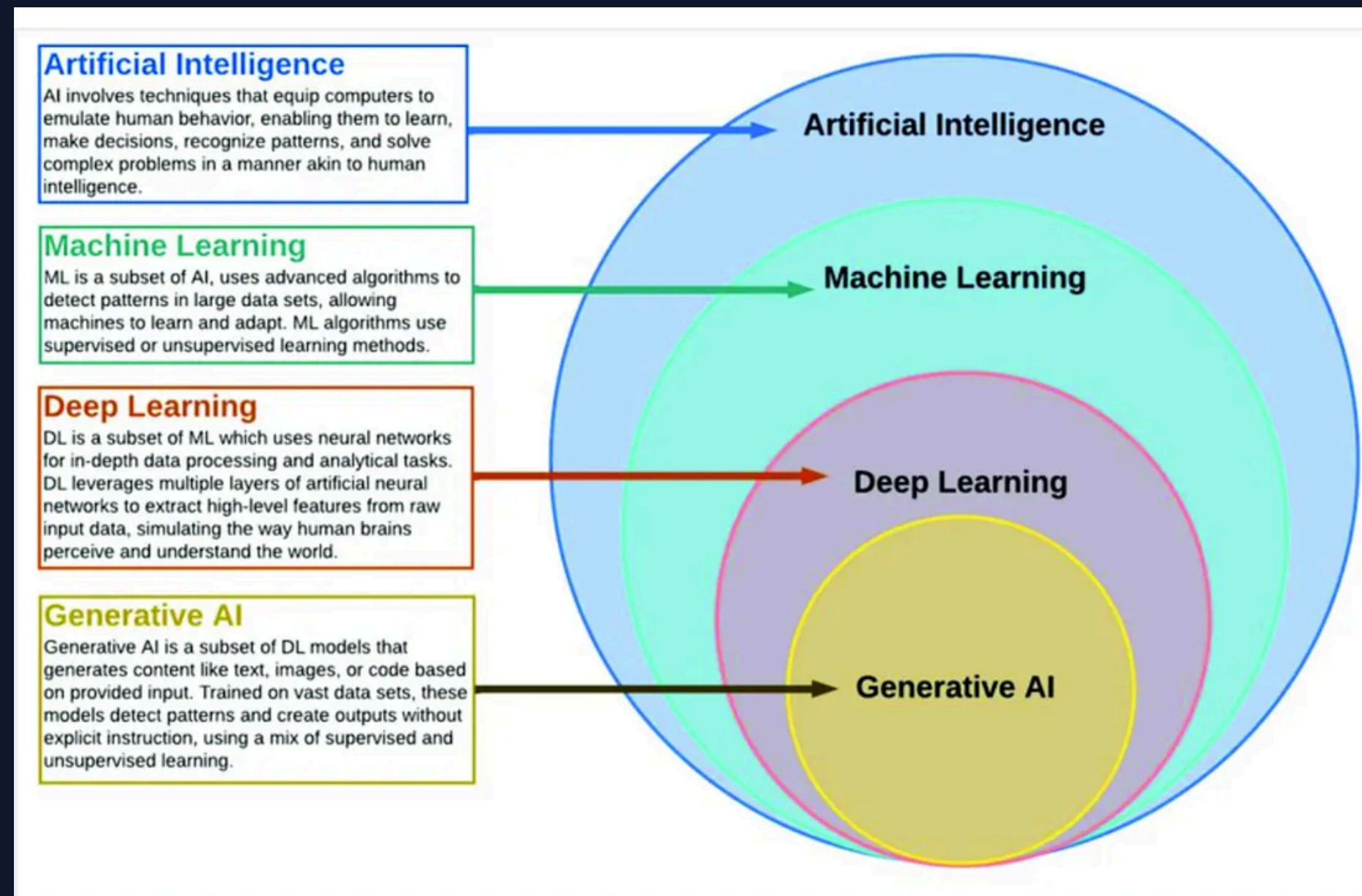
Local Mistral Demo (Hands-on)

Classical ML vs Deep Learning vs Gen AI

Classical ML		Deep Learning		Generative AI	
ERA	1950s – 2010s	ERA	2012 – 2020	ERA	2020 – Present
HOW	Hand-crafted features + statistical algorithms	HOW	Multi-layer neural nets learn features automatically	HOW	Foundation models trained on internet-scale data
EXAMPLES	Linear Regression, SVM, Decision Trees, Random Forest	EXAMPLES	CNNs, RNNs, ResNet, BERT	EXAMPLES	GPT-4, Gemini, DALL-E, Stable Diffusion
STRENGTH	Interpretable, works on small data	STRENGTH	Superhuman vision, speech, NLP at scale	STRENGTH	Creates text, images, code, audio — zero-shot
LIMIT	Feature engineering is manual & domain-specific	LIMIT	Needs huge labelled data; black box	LIMIT	Hallucinations, alignment, cost

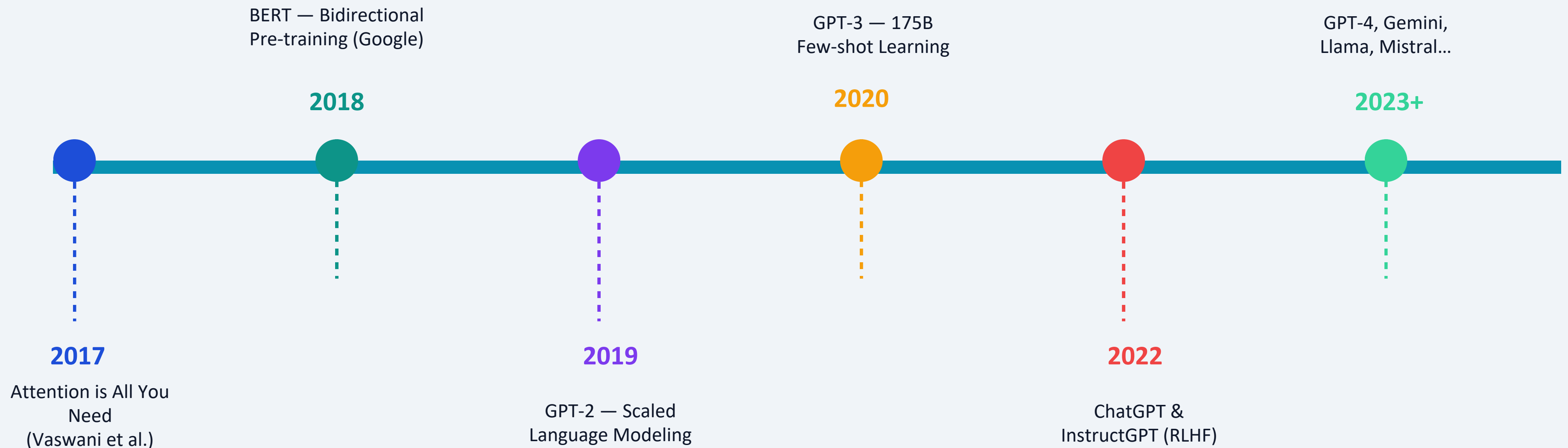
Each era builds on the last — Gen AI would not exist without DL breakthroughs.

Classical ML vs Deep Learning vs Gen AI



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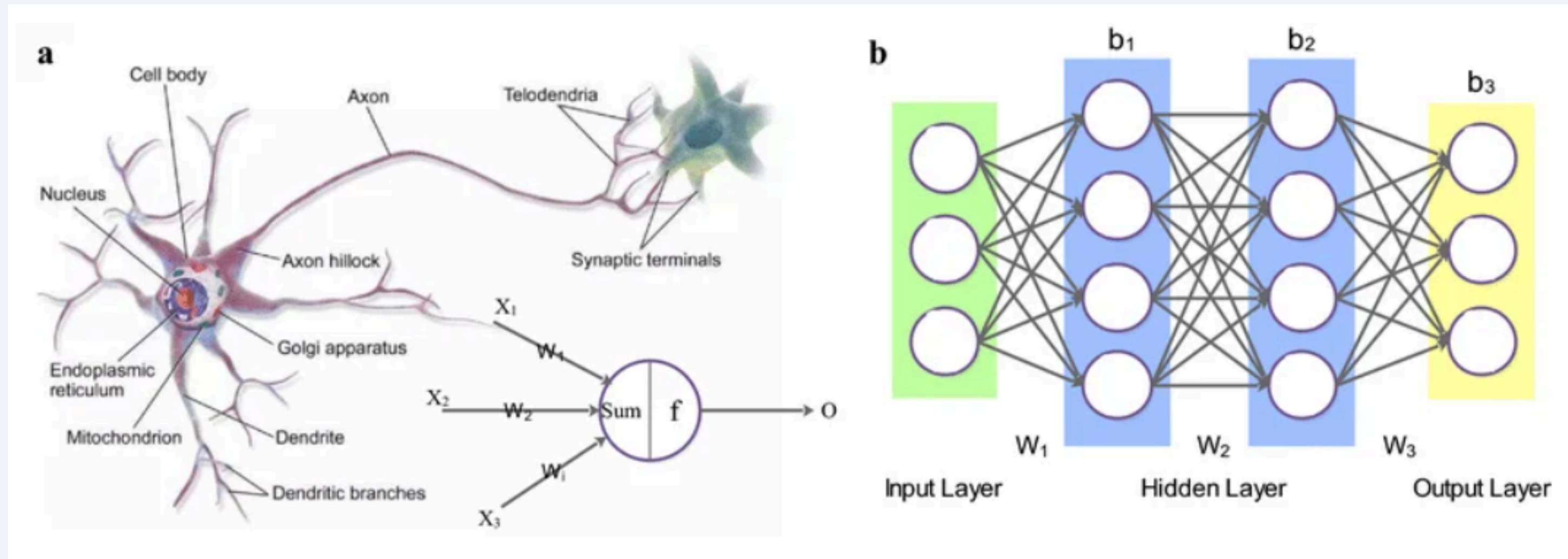
Transformer Architecture & Timeline



Key Transformer Concepts:

Self-Attention • Multi-Head Attention • Positional Encoding • Feed-Forward Layers • Encoder–Decoder or Decoder-only • Tokenisation • Embeddings

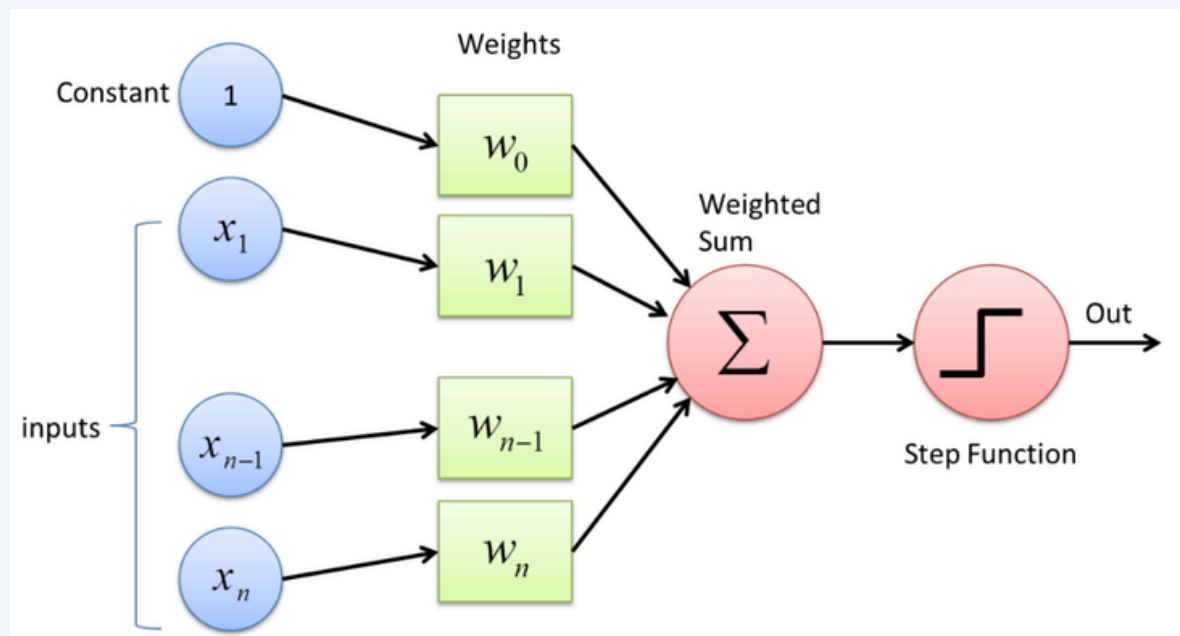
The Building Block - What is Neuron?



From Single Neuron → Deep Network (Perceptron to MLP)

1958 — The Perceptron

Frank Rosenblatt, Cornell



CAN DO:

Linear binary classification (spam / not spam, 0 / 1)

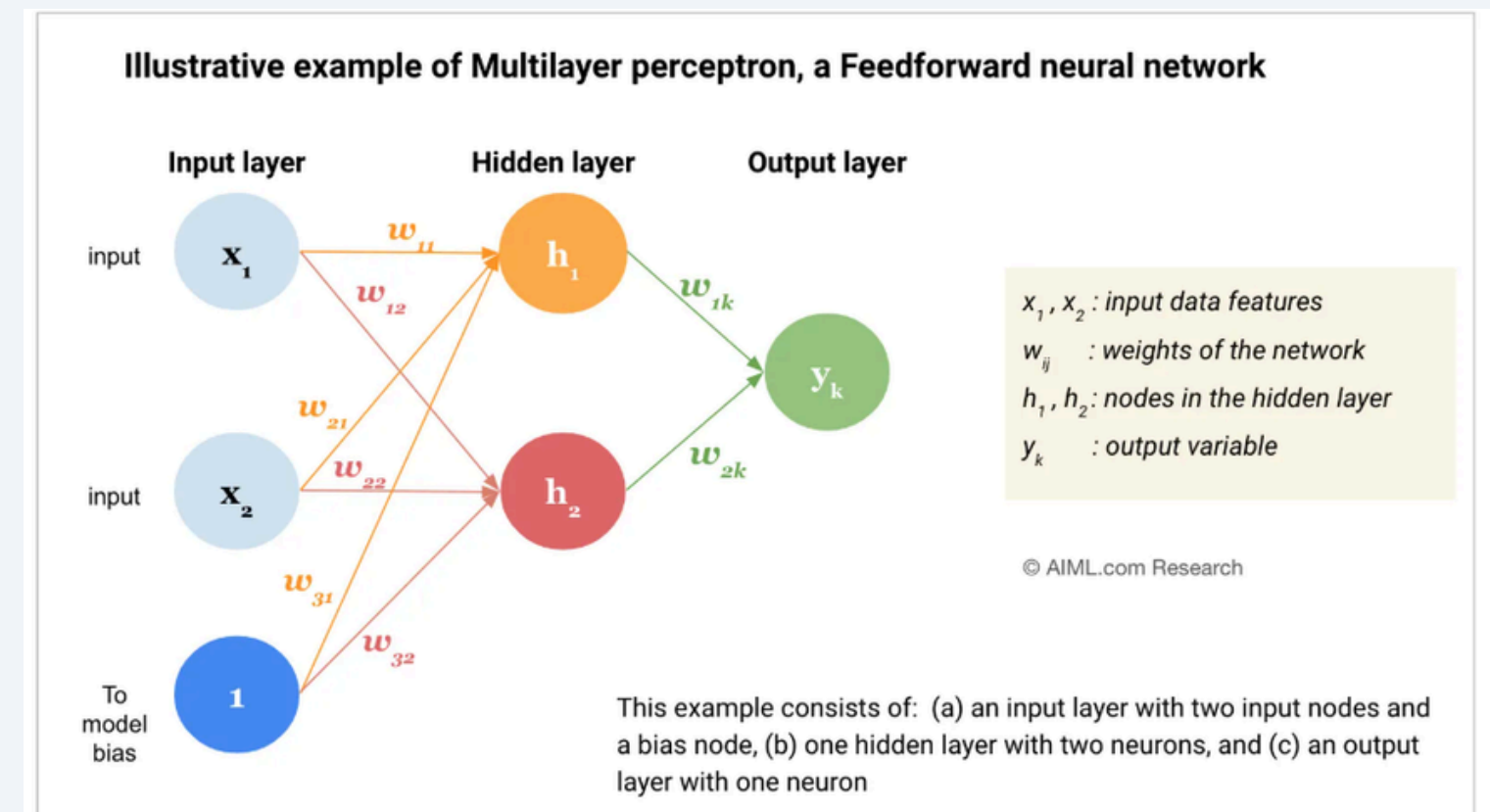
CANNOT DO:

XOR problem — any non-linearly separable data

→ Led to the first 'AI Winter' (1969–1980)

1986 — Multi-Layer Perceptron + Backprop

Rumelhart, Hinton & Williams — learns hidden representations



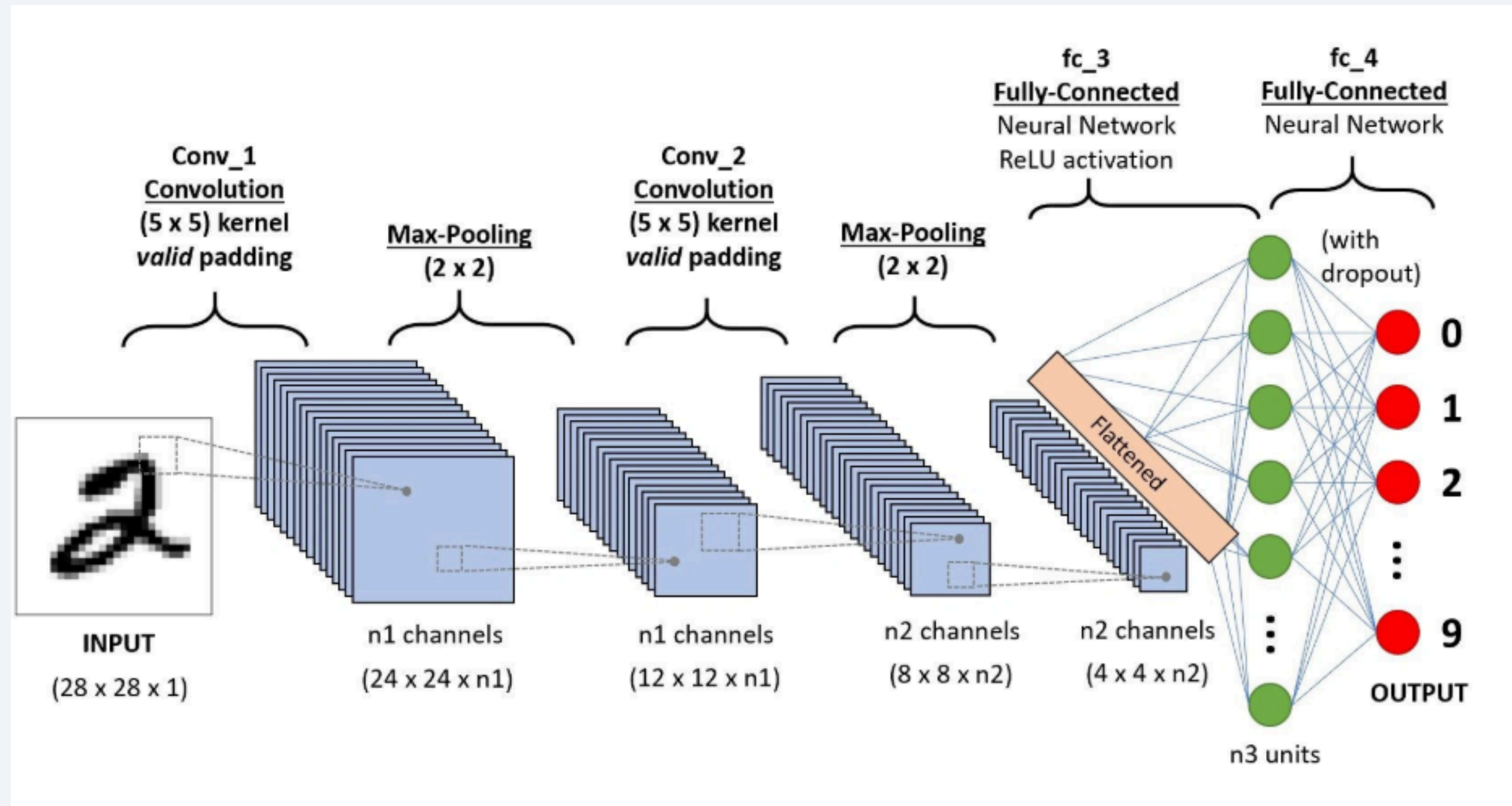
Backprop: error flows backward, adjusting all weights at once

Hidden layers learn abstract features the programmer didn't specify

More layers = more complex patterns, but training was unstable pre-2012

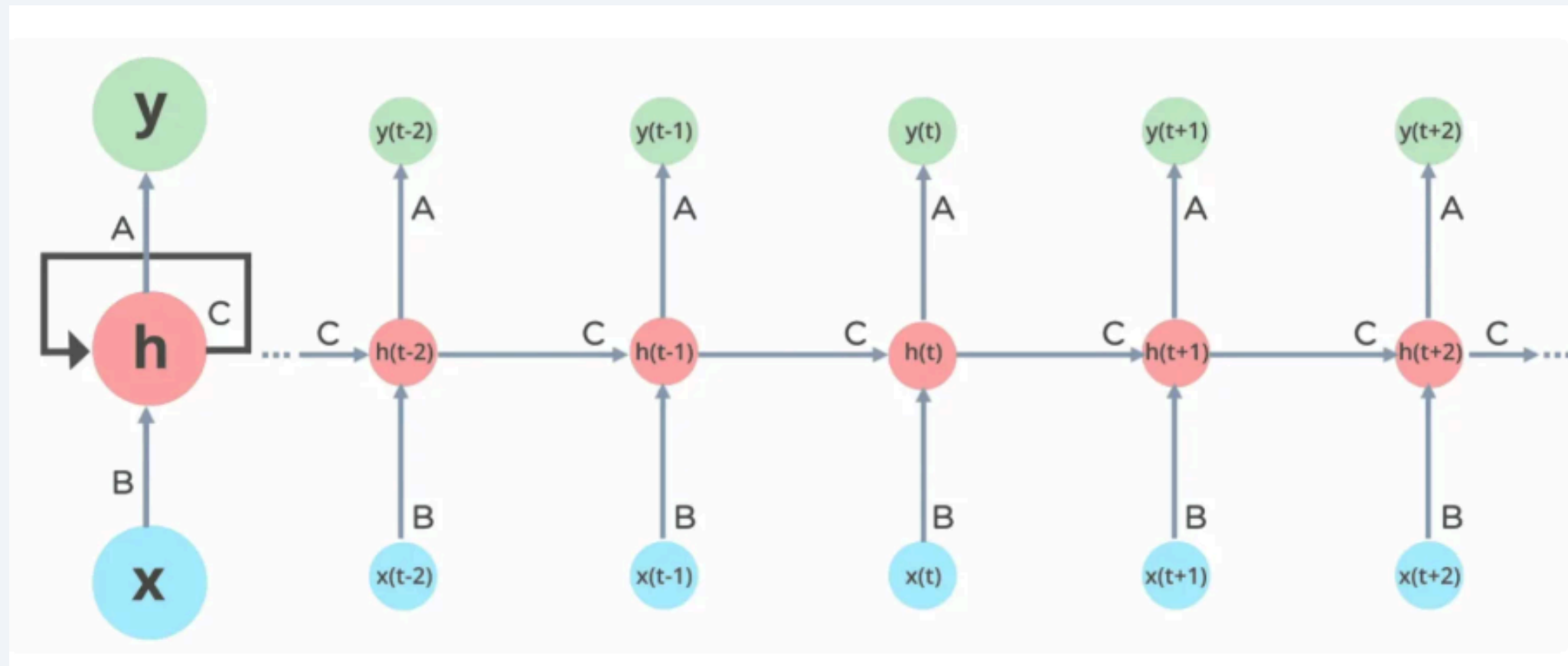
Convolutional Neural Networks (CNN) — 1998 / 2012

Designed for: Images, Video, Spatial Data — Key Insight: Local patterns repeat across an image



Recurrent Neural Networks (RNN) & LSTM — Sequences in Time

Designed for: Text, Speech, Time Series — Key Insight: Order matters; past context informs future



The Vanishing Gradient Problem

As sequences get long, gradients shrink exponentially — the network forgets earlier context.

Example: "The country where I grew up...still has a language that I..." — by 'language', it has forgotten 'country'.

LSTM — Long Short-Term Memory (1997)

-  **Forget Gate:** Decides what past info to discard
-  **Input Gate:** Decides what new info to store
-  **Output Gate:** Decides what to output now

Powered: Google Translate pre-2017, Siri speech recognition, Gmail Smart Compose.

The Evolution — And Why We Needed Something New

1958	1986	1998	1997	2017
Perceptron	MLP + Backprop	CNN	RNN / LSTM	Transformer
Best for:	Best for:	Best for:	Best for:	Best for:
Binary Classification	Tabular Data, Simple Patterns	Images, Video, Spatial	Text, Speech, Time Series	Everything — Text, Image, Audio
Simple, fast	Non-linear learning	Spatial feature sharing	Sequential memory	Parallel + global attention
Only linear problems	Unstable, slow to train	Poor at sequences/order	Slow, forgets long context	Needs large compute

The Transformer doesn't process left-to-right or neighbourhood-by-neighbourhood. Every token attends to every other token — simultaneously. That changes everything. → Next: How it works.

Diffusion vs Autoregressive Models

Autoregressive

Generate one token at a time

- ▶ Predict next token given all previous
- ▶ Training: causal language modelling
- ▶ Excellent for text, code, structured data
- ▶ Examples: GPT-4, Gemini, Llama, Mistral
- ▶ Fast inference; scales with compute
- ▶ Sampling: greedy, top-k, temperature

VS

Diffusion

Denoise from noise to signal

- ▶ Add noise in training; learn to reverse
- ▶ Iterative denoising at inference (50–1000 steps)
- ▶ Superb for images, audio, video
- ▶ Examples: Stable Diffusion, DALL-E 3, Sora
- ▶ Slow inference; parallelisable training
- ▶ Guidance: CFG, ControlNet, LoRA

Insight: Multimodal models (GPT-4o, Gemini Ultra) combine both paradigms — autoregressive for text, diffusion for images.

Open vs Closed Model Licences

Closed / Proprietary

Examples: OpenAI (GPT-4), Google (Gemini Ultra), Anthropic (Claude)

Strengths

- Best benchmark performance
- Managed safety & alignment
- Easy API access
- No infra setup needed

Considerations

- Paid per token — costs scale
- No weights access or fine-tuning
- Data sent to third-party servers
- Vendor lock-in risk

Open Weights

Examples: Meta (Llama 3), Mistral AI, Google (Gemma), TII (Falcon)

Strengths

- Run locally — full data control
- Fine-tune on proprietary data
- No per-token cost (infra cost only)
- Customise for domain tasks

Considerations

- Self-managed infra & scaling
- Smaller models lag closed frontier
- Licence nuances vary (commercial?)
- Responsible deployment on you

Hugging Face Model Hub

The GitHub of AI — 900,000+ models, 200,000+ datasets, 350,000+ spaces

Models

Filter by task (text-gen, image-class, ASR...), framework (PyTorch/TF/JAX), licence, language. One-click download.

Datasets

Versioned datasets with data cards — MMLU, SQuAD, Common Crawl, LAION. Hugging Face datasets library.

Spaces

Deploy interactive Gradio/Streamlit demos on HF infra. Zero-infra prototype sharing.

Transformers

`from_pretrained()` loads any model in 2 lines. Supports 100+ architectures unified under one API.

Model Cards

Standardised docs: intended use, training data, biases, evaluation, CO2 footprint.

Inference API

Serverless hosted inference for most models. Free tier + dedicated endpoints for production.

Local Mistral Demo — Hands-On

1

Install Ollama

```
curl -fsSL https://ollama.com/install.sh | sh
```

2

Pull Mistral 7B

```
ollama pull mistral
```

3

Chat in Terminal

```
ollama run mistral
```

4

Python API

```
import ollama
res = ollama.chat(model='mistral',
  messages=[{'role':'user',
    'content':'Explain RAG in 3 bullets'}])
print(res['message']['content'])
```

Why Mistral 7B?

Architecture

Decoder-only Transformer

Params

7 Billion (fits 8 GB VRAM)

Context

8,192 tokens

Techniques

Sliding Window Attn + GQA

Licence

Apache 2.0 — fully open

Speed

~40 tokens/sec on M2 MacBook

Variants

Mistral-7B-Instruct, Mixtral 8x7B

Ollama size

~4 GB (Q4 quantised)

Key Takeaways — What You Now Know

The AI Stack

Classical ML → DL → Foundation Models. Each layer adds generality but demands more data and compute.

Transformers Dominate

Self-attention + scale is the recipe. Almost every frontier model (text, image, audio) is transformer-based.

Two Generation Paradigms

Autoregressive (next token) for language; Diffusion (denoise) for images. Hybrids are emerging.

Licence Matters

Closed APIs = fast start, vendor risk. Open weights = control, privacy, cost — but you own the infra.

HF Hub is Your Toolbox

900k+ models, unified API, model cards. First stop for any open-source model exploration.

Local LLMs are Real

Mistral 7B via Ollama runs on a laptop. Local inference = privacy + zero per-token cost.

Continue Your Journey

Read

- Attention Is All You Need (Vaswani 2017)
- The Illustrated Transformer — Jay Alammar
- Andrej Karpathy: Let's build GPT

Learn

- fast.ai Practical Deep Learning
- DeepLearning.AI Generative AI Specialisation
- HF NLP Course (free)

Build

- Ollama + Open WebUI (local ChatGPT UI)
- LangChain / LlamaIndex for RAG
- Gradio — share a demo in 10 lines

Follow

- Papers With Code — trending models
- HuggingFace Blog — weekly releases
- Latent Space Podcast